STUDENT RECORD SYSTEM

UNIT TESTING REPORT

Unit-level verification of isolated modules and database operations

Document	TEST-UNIT-01
Project	Student Record System
Course	Software Engineering Mini Project
Team	Team 11
Implementation	C++17 application with SQLite persistence
Report basis	Recorded project test results and coverage.xml evidence available in the project conversation
Report date	10 November 2026

Prepared as a final testing report for the project documentation package.

1. Purpose and Scope

This report records unit-level testing evidence for the Student Record System. It focuses on isolated permission checks, audit operations, academic database validation, SQLite operations, and separately recorded GPA and authentication test executions. Integration and full application workflows are reported separately in the System Validation Report.

2. Test Environment

Component	Recorded configuration
Language	C++17
Database	SQLite 3
Local environment	Windows with MSYS2 UCRT64
Compiler	GCC 16.2.0
CI platform	GitHub Actions (Ubuntu, macOS and Windows jobs)
Compiler options recorded in local results	-Wall -Wextra -Werror

3. Unit Test Execution Summary

The five standalone programs listed in the recorded test-results snapshot were compiled and executed successfully. Authentication and GPA test programs were also separately executed locally and their results were recorded in the project handoff.

Test ID / Program	Coverage of test	Recorded result
TC-RBAC — rbac_test.cpp	Permission checks for four roles; case-insensitive role names; unknown roles denied	PASS
TC-AUD — audit_test.cpp	Audit table initialization; event insertion; five-year retention	PASS
TC-ACD — academic_validation_test.cpp	Subject/grade initialization; constraints; valid and invalid academic data	PASS
TC-SQL — sqlite_test.cpp	In-memory SQLite connection; table creation; insert and retrieve student record	PASS
TC-PERF — performance_test.cpp	Student listing, student search and attendance-summary query benchmarks	PASS; detailed in performance section
Authentication test — auth_test.cpp	Authentication test suite	28 PASS, 0 FAIL (local execution)
GPA test — gpa_test.cpp	GPA/CGPA calculation test suite	19 PASS, 0 FAIL (local execution)

4. Detailed Test Coverage

4.1 Role-Based Access Control

• Verified permission checks for four roles.

• Verified case-insensitive role-name handling.

• Verified that unknown roles are denied permissions.

• Recorded result: PASS.

4.2 Audit Module

• Verified audit-table initialization.

• Verified audit-event insertion.

• Verified five-year audit-log retention behavior.

• Recorded result: PASS.

4.3 Academic Database Validation

• Verified subject and grade table initialization, including repeated initialization.

• Verified credits and semester metadata columns.

• Verified insertion of students and subjects.

• Verified duplicate subject-code rejection.

• Verified boundary marks of 0 and 100 are accepted.

• Verified negative marks, marks above 100, duplicate student-subject grades, and grades referencing nonexistent students or subjects are rejected.

• Recorded result: PASS.

4.4 SQLite Operations

• Verified successful in-memory database connection.

• Verified students-table creation.

• Verified student-record insertion and retrieval.

• Recorded result: PASS.

5. Result Summary

All listed standalone unit/database tests passed in the recorded results. The authentication and GPA totals are reported as local execution evidence and are not represented here as independently verified GitHub Actions job totals.

Test group	Outcome
RBAC	PASS
Audit	PASS
Academic database validation	PASS
SQLite operations	PASS
Authentication (local)	28 PASS / 0 FAIL
GPA (local)	19 PASS / 0 FAIL

6. Limitations

• This report does not claim that every application requirement has a dedicated unit test.

• Attendance and report workflows are end-to-end acceptance tests and are documented in the System Validation Report.

• The performance test program measures SQLite query execution time; it is not a complete application response-time measurement.

7. Conclusion

The recorded unit and database-level tests demonstrate successful verification of the listed RBAC, audit, academic validation, SQLite, authentication and GPA test cases. The reported results are limited to the test programs and executions identified in this document.